



MetroCare Hospital Analytics Dashboard

Power BI • PostgreSQL • Power Query • DAX

Interactive Business Intelligence Solution for Hospital Operations



PROJECT OVERVIEW

Developed an end-to-end Business Intelligence dashboard for MetroCare Hospital by integrating PostgreSQL data into Power BI. The solution transforms operational healthcare data into interactive visual analytics, enabling management to monitor revenue, patient demographics, appointments, doctor performance, and operational efficiency through a centralized reporting platform.



DATA QUALITY IMPROVEMENTS



Date Standardization

Converted all date/time fields into correct formats.



Data Type Corrections

Fixed decimal precision for Unit Price and Discount.



Data Cleaning

Removed invalid billing records (Quantity ≤ 0).



Standardization

Unified inconsistent Appointment Status and Payment Mode values.



Privacy Protection

Removed patient email and phone information.



Business Friendly Model

Renamed columns with meaningful business names.



Power Query Transformations

Split doctor names, created Item Revenue Summary, and merged Patient Category into Appointments.



DimDate & Star Schema

Built Date Dimension table and implemented one-to-many star schema relationships.



KEY DAX MEASURES

#	MEASURE NAME	DESCRIPTION / PURPOSE
1	Total Billed Revenue	Total hospital revenue after discounts.
2	Total Patients	Distinct count of registered patients.
3	Total Appointments	Distinct count of appointments.
4	Average Revenue	Average revenue generated per appointment.
5	Completed Appointments	Number of appointments successfully completed.
6	Cancelled %	Percentage of appointments that were cancelled.
7	No-Show %	Percentage of appointments that were no-shows.
8	Active Insurance Revenue	Revenue generated through Insurance Claims.
9	Self Pay Revenue	Revenue generated through direct (self) payments.
10	YTD Revenue	Year-to-date total revenue.
11	Previous Month Revenue	Total revenue for the previous month.
12	MoM Growth %	Month-over-month revenue growth percentage.

MetroCare Hospital Analytics

Total Billed
Revenue
10.51M

Total Patients
300

Cancelled
Appointment %
11.33

Total
Appointments
1K

Total Billed Revenue by Month



Total Billed Revenue by Department Name



DASHBOARD HIGHLIGHTS



TOTAL REVENUE
10.5M
PKR



TOTAL
APPOINTMENTS
1,000



TOTAL PATIENTS
300



CANCELLATION RATE
11.3%



Geographic Distribution

Patient distribution across cities and regions.



Doctor Performance

Top doctors by revenue, appointments & efficiency.



Department Analytics

Revenue and appointment trends by department.



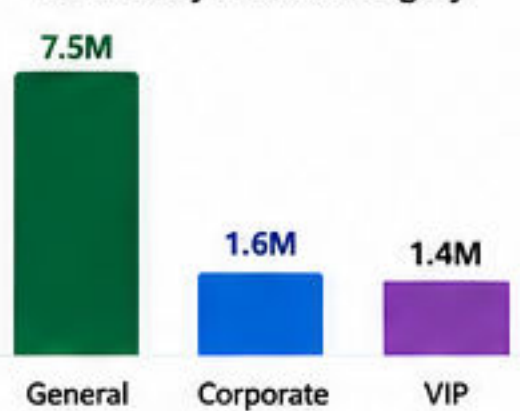
BUSINESS INSIGHT

Analysis of patient categories revealed that **General patients** generated approximately **7.5M** in revenue, significantly outperforming **Corporate (1.6M)** and **VIP (1.4M)** patients.



RECOMMENDATION: The hospital should prioritize improving services and retention strategies for General patients while exploring opportunities to increase high-value engagement within Corporate and VIP segments through targeted healthcare packages and partnerships.

Revenue by Patient Category



TECHNOLOGIES USED



PostgreSQL



Power Query



Power BI



DAX



Star Schema
Model



DELIVERABLES



Power BI Dashboard (.pbix)



Dashboard PDF Export



Project Summary (One Pager)

